



ENTERPRISE AI ENGINEERING

Building Trustworthy Enterprise RAG and AI Agents

From retrieval demos to measurable, governable, and dependable systems.

RAG

AI AGENTS

EVALUATION

GUARDRAILS

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A fluent answer is not enough.

Enterprise AI must handle fragmented knowledge, changing policy, permissions, and operational risk.

What the system must do

- **Retrieve** the right evidence within permission boundaries.
- **Decide** whether to answer, ask, abstain, refuse, escalate, or use a tool.
- **Expose** provenance, uncertainty, and failure paths.
- **Protect** private, unsafe, and stale information.
- **Measure** quality before adding architectural complexity.

The common demo

query -> retrieve -> generate

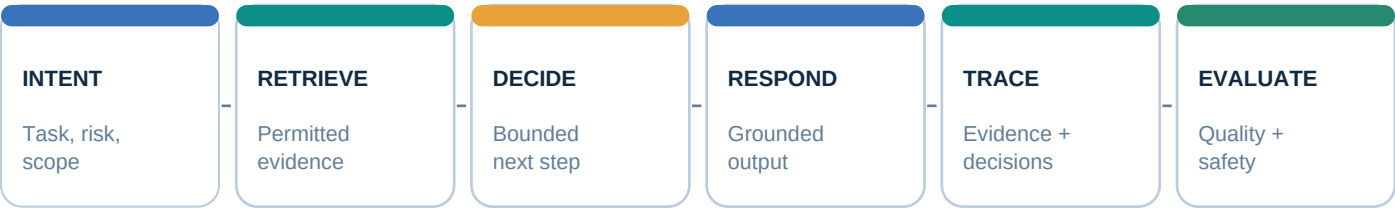
What is missing?

- Decision policy
- Permission scope
- Traceability
- Evaluation
- Fallback behavior

Design principle: RAG is not the product. It is the evidence layer inside an observable agent loop.

Make every important decision visible.

The loop connects evidence retrieval to bounded action, traceability, and continuous evaluation.



Decision taxonomy

- answer with evidence
- ask a clarifying question
- retrieve again or use a tool
- abstain or refuse
- escalate for human review

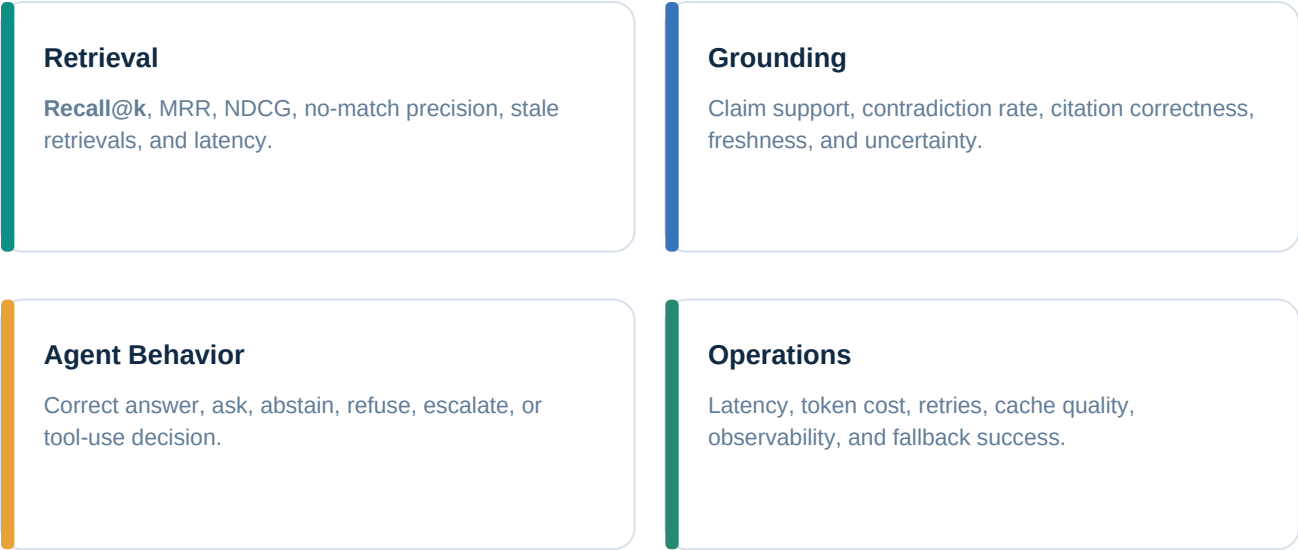
Trace contract

- intent and risk classification
- retrieved evidence and permissions
- decision and reason code
- model, prompt, latency, and cost
- safety and evaluation outcome

Outcome: When a result is wrong, the team can identify whether the failure came from intent classification, retrieval, decision policy, generation, or safety review.

Trust requires more than retrieval.

Quality, privacy, safety, and operations must be measured as separate system properties.



Governance sequence

- Authenticate and authorize.
- Determine allowed memory and knowledge scope.
- Retrieve only permitted evidence.
- Generate under versioned policy.
- Review, repair, or use a safe fallback.

Architecture escalation

- Start with a measurable baseline.
- Name the failure and target metric.
- Estimate latency, cost, privacy, and maintenance.
- Compare against a fixed evaluation set.
- Remove components that do not earn their place.

Separate demonstrated capability from active experiments.

Credibility comes from working systems, tests, explicit boundaries, and measurable next steps.

DEMONSTRATED

- **Mixed-format RAG:** Markdown, Excel, and Word ingestion with 487 indexed chunks, persistent vector retrieval, and grounded generation.
- **AI automation:** A nine-node workflow connecting CRM data, model reasoning, workflow updates, and notifications.
- **Avaloka AI:** Risk routing, planning, guardian review, repair, fallback, evidence-backed memory, diagnostics, and behavior evals.
- **Verification:** 77 Avaloka tests and 200 Auralith tests passed on June 8, 2026.

ACTIVE EXPERIMENTS

- **Normalize** complete agent traces across classification, retrieval, decision, generation, and review.
- **Benchmark** deterministic memory retrieval using Recall@5, MRR, NDCG@5, privacy checks, and latency.
- **Escalate** to embeddings, hybrid retrieval, reranking, RAPTOR, or GraphRAG only for measured failures.
- **Verify** final claims against permitted evidence, provenance, freshness, and uncertainty.

PROFESSIONAL FOCUS

Engineering leadership for AI systems that can be trusted in real use.

I bring together distributed systems, AWS and event-driven architecture, Applied AI, retrieval, guardrails, evaluation, and operational reliability.

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